

Omar Sharif

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Graduate Teaching Assistant and Researcher specializing in AI and machine learning. Experienced in building and evaluating PyTorch and TensorFlow models on imbalanced and healthcare datasets, including a deep learning approach for class imbalance mitigation and a blood glucose prediction system using continuous monitoring data. Proven ability to translate complex research problems into deployable solutions. Seeking to apply advanced AI expertise to drive innovative, data-driven outcomes in a research-focused or industry role.

Technical Skills

Programming languages: Python, SQL, C/C++, Java, HTML/CSS, Shell scripting, Slurm scripting.

Tools & Frameworks: Pytorch, Tensorflow, Keras, WandB, Github workflow, Tensorboard, PowerBI, numpy, scikit-learn, pandas, OpenCV, matplotlib.

Applications: Machine learning, deep learning, computer vision, Transformer Models, LLMs, Gen AI, Reinforcement Learning, NLP, ML libraries, ML Ops, object-oriented programming, data analysis, model evaluation, data visualization.

Projects and Awards

Graduate Research and Creative Scholarship Grant 2019

SAWTab: Smoothed Adaptive Weighting for Tabular Data in Semi-Supervised Learning. | [Paper](#) 2024

- Developed a semi-supervised learning method to improve minority-class prediction in imbalanced tabular datasets by designing an adaptive weighting strategy to generate higher-quality pseudo-labels for unlabeled data, achieving up to 85% prediction accuracy and outperforming strong baseline methods.

Analyzing Hematological Parameters in Diabetes: Toward a Better Understanding of Pathophysiology. 2025

- Analyzed an institutional cross-sectional dataset of 62 diabetic and 72 non-diabetic patients using correlation matrices with significance threshold $p < 0.05$.

A Literature Review on Emotion Recognition using various methods. | [Paper](#) 2017

- Conducted a comprehensive review of 20+ peer-reviewed studies on emotion recognition, comparing deep learning architectures, feature representations, and performance trends. published in a peer-reviewed journal with 31+ citations.

Blood Glucose Prediction using Continuous Glucose Monitoring data. 2025

- Analyzed 10+ transformer- and RNN-based deep learning models for blood glucose prediction using CGM data, highlighting accuracy limitations, data challenges, and opportunities for model improvement.

Experience

Graduate Teaching Assistant and Researcher Old Dominion University
Department of Computer Science January 2022 – December 2025

- Led 10+ Java/C++ lab sessions per semester for 50+ students, held weekly office hours, and graded 300+ assignments with 95% course pass rate.
- Designed 12+ assignments and 20+ exercises for Java and Discrete Structures courses, used by 200+ students across multiple semesters.
- Developed several deep learning projects on imbalance on tabular data and time-series data, mitigating class imbalance and improving glucose prediction accuracy by 2-3%.

Graduate Assistant University of South Dakota
Department of Computer Science August 2018 – December 2019

- Programmed the Parallax Activity Bot 360 in C++ to solve mazes and perform autonomous parking, enabling the robot to navigate complex layouts without manual control and increasing successful movement tasks by 30%
- Engineered advanced face detection and speech recognition modules using OpenCV and C++, resulting in a real-time response rate improvement of 20% for the EZ Robot JD's interactive functionalities.

Education

Old Dominion University Norfolk, VA
Master of Science in Computer Science January 2022- May 2026(Expected)

University of South Dakota Vermillion, SD
Master of Science in Computer Science August 2018- August 2021

American International University- Bangladesh Dhaka, Bangladesh
B.S. in Computer Science and Engineering May 2010 - April 2015